

SCHOOL OF COMPUTER AND SYSTEMS SCIENCES
JAWAHARLAL NEHRU UNIVERSITY
New Delhi -110067

MCA III SEMESTER

Internal Assessment I (Monsoon) CS 425 Blockchain Technology

Date: 17-09-2024 & Time: 3:00 PM to 5:00 PM

Max Marks: 50

Question No. 1 Carry one mark each (Answer ALL)

(1x5=5)

1a) What is an advantage of a public blockchain?

- A. It does not use disinterested third parties to secure blocks, as all participants have a vested interest.
- B. It is more resilient against fraud, because it uses federated nodes to combat fraud.
- C. It is open to everyone in the world without permission and licensing requirements.
- D. Its networks are built by for-profit companies and the working of the network is guaranteed

1b) What is the task of miners in a blockchain network?

- A. Miners act as a single third party to aggregate records and provide trust in the network by the miners.
- B. Miners are computers that allow access to the blockchain, ensuring corrupt nodes will stay low.
- C. Miners are nodes that compete for a reward by calculating the correct nonce to make a transaction.
- D. Miners determine the consensus rules that should be followed and interfere when these rules are broken.

1c) Which characteristic of a blockchain network is also its protection?

- A. The greater the number of full independent nodes, the harder it is to compromise the data in blockchain.
- B. The lower the number of miners in the blockchain, the higher the incentive is for securing the network.
- C. The more centralized the control of the blockchain is, the harder it is to secure the data and avoid fraud.
- D. The more complicated the Proof of Work algorithm is, the more rewarding it is to secure the network

1d) An organization wants to develop smart contracts, based on blockchain technology. The organization does not wish to burden employees with maintaining the security of the blockchain. What blockchain technology fits the organization best?

- A. A hybrid blockchain
- B. A private blockchain
- C. A public blockchain
- D. A Consortium blockchain

1e) What is digital fiat currency?

- A. A digital form of currency, that represents a country's financial reserves
- B. An e-currency, that creates a transparent and borderless debt market
- C. An online system, that enables making transactions without a bank account
- D. A cryptocurrency for digital transaction

Question No. 2 Carry two marks each (Answer ALL)

(2x5=10)

- 2a) Define Genesis block
- 2b) State how CAP Theorem can be achieved in blockchain
- 2c) Why transaction fess is required in blockchain
- 2d) Define Bitcoin
- 2e) Characterize the role of full node in blockchain.

Question No. 3 to 10 carry five marks each (Answer any 7 questions)

(5x7 =35)

- 3. Describe the features of a blockchain
- 4. Differentiate centralized, decentralized and distributed systems.
- 5. Construct a merkle tree using the hash function 'key mod 5' for the following input 364, 225, 618, 992, 548, 28.
- 6. Describe the layers in the Blockchain.
- 7. Illustrate the bitcoin transaction life cycle with a suitable example
- 8. Discuss the types of wallets deployed in bitcoin network.
- 9. Sketch the structure of a block and tabulate the 80 bytes distribution of the block header.
- 10. Summarize the process of a transaction creation and how it is added to a block and to a chain.

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MCA III SEMESTER

Internal Assessment II (Monsoon) CS 425 Blockchain Technology

Date: 03-12-2024 & Time: 9:00 PM to 11:00 PM

Max Marks: 50

Question No. 1 to 5 Carry two marks each (Answer ALL)

(2x5=10)

1. List the properties of hash functions.
2. Define ECDSA.
3. Define Wei.
4. State the Elliptic Curve equation.
5. What is gas fee.

Question No. 6 and 7 carry five marks each

(5x2 =10)

6. State the features of cryptography.
7. Discuss the two types of accounts in Ethereum.

Question No. 8 and 9 carry fifteen marks each

(15x2 =30)

8. Explain SHA256 algorithm with output at various steps for the input “hello world”.
9. Determine the points on the curve $y^2 = x^3 + ax + b \pmod{11}$, for set of all integer and show that point (3,8) falls on the curve.
